

Feature Engineering Interview Questions & Answers

Beginner → Advanced | Data Analyst / Data Scientist / ML Internship

Quick Definition

Feature engineering is the process of creating, transforming, selecting, or extracting useful features from raw data so that machine-learning models can learn patterns more effectively.

1. Basic Feature Engineering Questions

Q1. What is Feature Engineering?

Answer: Feature engineering is the process of transforming raw data into meaningful features that can improve a machine-learning model's ability to learn patterns.

Q2. Why do we perform feature engineering?

Answer: To make useful information available to the model, improve predictive performance, reduce noise, represent data in a suitable form, and sometimes reduce model complexity.

Q3. What is a feature?

Answer: A feature is an input variable used by a machine-learning model to make predictions. For example, age, income, tenure, or transaction count can be features.

Q4. What is the difference between feature engineering and feature selection?

Answer: Feature engineering creates or transforms features. Feature selection chooses the most useful features from the available set and removes irrelevant or redundant ones.

Q5. What is the difference between EDA and feature engineering?

Answer: EDA helps us understand the data and discover patterns, quality issues, and relationships. Feature engineering uses those findings to create or transform model-ready features.

Q6. When should feature engineering be performed?

Answer: It should be planned after understanding the problem and data, but all transformations must be fitted only on training data when they learn parameters from data, to avoid leakage.

2. Numerical Feature Engineering

Q1. What is feature scaling?

Answer: Feature scaling puts numerical features on comparable scales. Common methods include StandardScaler and MinMaxScaler.

Q2. Standardization vs normalization?

Answer: Standardization typically transforms a feature to mean 0 and standard deviation 1. Min-max normalization commonly maps values to a fixed range such as 0 to 1.

Q3. When is scaling important?

Answer: It is especially important for distance- or gradient-based algorithms such as KNN, K-Means, SVM, and many linear/neural-network models. Tree-based models generally do not require scaling.

Q4. What is log transformation?

Answer: A log transformation compresses large values and can reduce right skew. It is useful for suitable positive, highly skewed variables such as income or transaction amounts.

Q5. What is binning?

Answer: Binning converts a continuous numerical variable into intervals or categories, such as age groups. It can simplify nonlinear relationships but may lose information.

Q6. How do you handle outliers during feature engineering?

Answer: First investigate whether they are valid. Depending on the problem, you can retain them, cap/winsorize them, transform the feature, or use robust methods. Do not remove them automatically.

3. Categorical Feature Engineering

Q1. What is categorical encoding?

Answer: Categorical encoding converts categories into numerical representations that a machine-learning model can use.

Q2. What is One-Hot Encoding?

Answer: It creates a separate binary column for each category. It is commonly used for low-cardinality nominal variables.

Q3. What is label encoding?

Answer: It maps categories to integer values. It is appropriate when the categories have meaningful order, but arbitrary integer codes can be misleading for nominal categories.

Q4. What is ordinal encoding?

Answer: It maps ordered categories according to their natural rank, such as Low < Medium < High.

Q5. What is target encoding?

Answer: It replaces a category with a statistic derived from the target, often the target mean. It must be implemented carefully with proper cross-validation or training-only calculations to prevent target leakage.

Q6. What is one problem with high-cardinality categorical variables?

Answer: One-hot encoding can create a very large number of columns. Alternatives include frequency encoding, carefully regularized target encoding, hashing, or grouping rare categories, depending on the problem.

4. Feature Transformation & Creation

Q1. What is a derived feature?

Answer: A derived feature is created from one or more existing variables, such as BMI from weight and height or total amount from price × quantity.

Q2. What are interaction features?

Answer: Interaction features represent the combined effect of variables, such as income × age. They can help models capture relationships that simple individual features may miss.

Q3. What is polynomial feature engineering?

Answer: Polynomial features add powers and interactions such as x^2 , x^3 , or $x_1 \times x_2$ to help suitable models represent nonlinear relationships.

Q4. How do you engineer date/time features?

Answer: Extract meaningful components such as year, month, day, day of week, hour, weekend indicator, elapsed time, or time since an event. The choice depends on the business problem.

Q5. How do you engineer text features?

Answer: Common approaches include word/character n-grams, TF-IDF, counts, text length, keyword indicators, and embeddings. The choice depends on the model and task.

Q6. How do you create ratio features?

Answer: Divide one meaningful quantity by another, such as debt-to-income or sales-per-customer, while handling zero denominators and checking whether the ratio makes business sense.

5. Feature Selection

Q1. What is feature selection?

Answer: Feature selection is the process of keeping useful features and removing irrelevant, redundant, or harmful features.

Q2. What are common feature selection methods?

Answer: Filter methods use statistics or correlation; wrapper methods evaluate feature subsets using a model; embedded methods perform selection during training, such as L1 regularization or tree-based importance.

Q3. What is correlation-based feature selection?

Answer: Highly correlated predictors can contain redundant information. You can inspect a correlation matrix and remove one of strongly redundant features when justified, while considering the target and model.

Q4. What is Recursive Feature Elimination (RFE)?

Answer: RFE repeatedly trains a model, ranks features, removes the least important ones, and continues until the desired number of features remains.

Q5. What is dimensionality reduction?

Answer: It transforms many features into a smaller set of components. PCA is a common example. Unlike feature selection, the resulting components may not correspond directly to the original features.

6. Data Leakage & Practical Interview Questions

Q1. What is data leakage in feature engineering?

Answer: Data leakage occurs when information that would not be available at prediction time enters the training features or preprocessing process. It can produce unrealistically strong validation results.

Q2. Give an example of feature leakage.

Answer: If predicting whether a customer will default, using a field created after the default event—such as a recovery amount recorded later—would leak future information.

Q3. How do you prevent leakage during scaling or imputation?

Answer: Split the data first. Fit the scaler or imputer on the training set only, then use that fitted transformation on validation/test data. Pipelines help enforce this workflow.

Q4. How do you decide whether to create a new feature?

Answer: Start from business/domain knowledge and EDA findings. Ask whether the feature is available at prediction time, represents useful information, is reliable, and improves validation performance.

Q5. What if feature engineering increases the number of features too much?

Answer: Check redundancy, remove irrelevant features, use feature selection or dimensionality reduction, regularize the model, and validate whether the added features actually improve performance.

Q6. How do you evaluate a new engineered feature?

Answer: Compare a baseline model with and without the feature using a proper train/validation or cross-validation setup and an appropriate metric. Also check stability, leakage, interpretability, and business usefulness.

7. Practical Workflow

Q1. Explain your feature engineering workflow in an interview.

Answer: Understand the business problem and target → split data appropriately → inspect data types and distributions → handle missing values → encode categorical variables → transform numerical variables when needed → create domain-based features → select/reduce features → build preprocessing in a pipeline → validate without leakage.

Q2. What would you do with a highly skewed numerical feature?

Answer: Investigate the cause and whether the skew is meaningful. Depending on the model and context, consider log/power transformation, robust scaling, binning, or leaving it unchanged if the model handles it well.

Q3. How would you handle a categorical column with rare categories?

Answer: Group very rare categories into an 'Other' category when appropriate, or use an encoding method suited to high cardinality. The grouping/encoding decision should be learned from training data.

Q4. What is the most important rule in feature engineering?

Answer: The engineered feature must be available at prediction time and the transformation must be learned without using validation/test information. In other words: avoid leakage.

Useful Python Feature Engineering Commands

Purpose	Python
Create a new column	<code>df['new'] = df['a'] * df['b']</code>
Log transformation	<code>df['log_x'] = np.log1p(df['x'])</code>
One-hot encoding	<code>pd.get_dummies(df, columns=['cat'])</code>
Value counts	<code>df['column'].value_counts()</code>
Unique values	<code>df['column'].nunique()</code>
Standard scaling	<code>StandardScaler()</code>
Min-max scaling	<code>MinMaxScaler()</code>
Train/test split	<code>train_test_split(X, y, test_size=0.2)</code>
Column-wise preprocessing	<code>ColumnTransformer([...])</code>
Pipeline	<code>Pipeline([...])</code>
Polynomial features	<code>PolynomialFeatures(degree=2)</code>
Feature selection	<code>SelectKBest(...)</code> / <code>RFE(...)</code>

Interview Tip

Do not memorize only definitions. Explain the reason behind your choice. A strong answer sounds like: "I checked the feature distribution and cardinality, selected an appropriate transformation/encoding, fit the preprocessing only on the training data, and verified the impact using cross-validation without leakage."